

Towards Design of Airfoil Pressure Tap Locations for Real-Time Predictions Under Uncertainty Using Bayesian Neural Networks

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Flow separation can be highly dangerous for aircraft operations especially during takeoffs and landings, and active flow control is an important research area tasked with developing technologies to delay the effects of flow separation. Designing active controllers requires reliable real-time in-flight detection of aerodynamic characteristics from sensor measurements. In this work, we focus on (a) constructing a mapping from sparse and noisy pressure tap readings to aerodynamic quantities of interest using Bayesian neural networks (BNNs) trained from SU2 flow simulations; and (b) designing pressure tap location configurations by assessing the BNN performance on testing data with an uncertainty-incorporated loss criterion. BNNs are a type of machine learning model that can provide rapid online predictions as well as uncertainty information surrounding the predicted values, and thus highly useful for supporting decision-making that may impact aircraft and passenger safety. This paper presents a proof-of-concept study of a real-time flow separation detection tool in a simplified two-dimensional environment.

I. Introduction

Flow separation is an important consideration for aircraft design, since it can lead to decreased lift and increased drag especially at low airspeeds and high angles of attack. This is particularly dangerous for takeoff and landing operations where an aircraft, already close to the ground, may stall and rapidly lose altitude. The area of flow control research thus develops techniques and technologies to delay the onset of flow separation [1, 2]. Passive (predetermined) control devices such as vortex generators [3, 4] do not require sensors or knowledge of the flight condition, but tend to be inefficient or can introduce adverse effects such as increased parasitic drag. In contrast, active (reactive) control methods such as blowing and suction [5] and synthetic jets [6, 7] operate only when necessary and may adapt depending on the current flight condition, and thus can be substantially more energy efficient. While active separation control has been investigated for a long time, it has not been widely deployed, and stall warning systems today still primarily rely on conservative airspeed and angle of attack indications. Challenges preventing a wide adoption of active separation control include the requirements for reliable sensor readings, fast and accurate state estimations together with uncertainty quantification, and robust closed-loop control algorithms.

Our present work focuses on estimating aerodynamic quantities of interest (QoIs) from pressure tap readings (e.g., via [8]), which can provide valuable information to an active separation controller. With often only sparse and noisy pressure measurements available, one typically needs to infer the underlying flight conditions and compute the QoIs through the governing physics, usually in the form of nonlinear partial differential equations (PDEs). Solving the underlying inverse problem involves repeated flow simulations, each highly expensive and can take hours or longer even on a computer cluster. An online deployment of such procedure, especially coupled with high-fidelity modeling and scale-resolving numerics, would be intractable. We thus seek accelerations through machine learning (ML) techniques, and build data-driven models offline on the ground leveraging high-performance computing resources and physics-based simulations, and then utilize the resulting ML models for rapid online predictions in-flight. Our vision is to eventually develop a ML model that can be integrated into a real-time, closed-loop controller to perform reliable active flow control.

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In this study, we investigate one particular ML model form: neural networks (NNs) [9]. NNs are a powerful tool with its high expressivity, and abilities to handle large datasets and to discover useful features. However, adoption of NNs, or any other ML models, necessitates the understanding of uncertainty in their predictions, especially when these predictions are used for high consequence decisions that concern the safety of an aircraft and its passengers. In contrast to classical NNs that only provide single-value predictions, Bayesian neural networks (BNNs) [10–12] are capable of producing a prediction as well as its uncertainty distribution. BNN thus allows the use of control algorithms that can take into account the uncertainty information provided by the BNN—for instance, if the uncertainty interval on a prediction is too large, the controller may decide to refrain from making a control input based on that unreliable prediction or to prompt the pilot for intervention. We will use BNNs in this paper to enable ML predictions with uncertainty. The main objectives and contributions of our paper are:

- to adopt BNNs and enable rapid predictions of aerodynamic performance QoIs (with quantified uncertainty) given sparse and noisy pressure tap readings; and
- to assess BNN performance under different pressure tap location configurations and explore an initial design under uncertainty.

We note that this paper presents a proof-of-concept illustration of an in-flight flow separation detection tool where the flow simulations are based on the two-dimensional Reynolds-averaged Navier-Stokes (RANS) equations—it is a simplified representation of the real life situation. Indeed, for tapered and swept wings used in modern aviation, the flow over the wings is very much three-dimensional. Additionally, it is well understood that RANS produces inaccurate predictions for separated or mildly separated flows, and accurate predictions require scale-resolving methods such as large eddy simulations. However, such computations are much more expensive, making the generation of sufficient data points for our current investigation intractable. They are therefore outside the scope of this paper.

The paper is organized as follows. Section II describes the problem formulation and application background. We then describe the numerical solver and ML framework in Sec. III. Results are presented in Sec. IV, followed by concluding remarks in Sec. V.

II. Problem Formulation

Our first goal is to achieve real-time predictions of aerodynamic performance QoIs $y := \{C_L, C_D, C_{M_z}, x_s\}$ (lift coefficient, drag coefficient, moment coefficient, and flow separation location) given noisy pressure readings $x := \{c_{p,i}\}_{i=1}^{n_p}$ from n_p sparse surface pressure taps on the upper surface of an airfoil. As a proof-of-concept demonstration of the overall framework, we use the NACA4421 airfoil (Fig. 1(a)) and focus on steady two-dimensional flows. We assume the underlying flight conditions $\theta := \{M_\infty, \text{Re}, \alpha\}$ (free stream Mach number, Reynolds number, and angle of attack) are unknown and not directly observable. We further assume access to a computational fluid dynamics (CFD) solver that can predict the QoIs and pressure tap readings given flight conditions via the following relationships:

$$x = f(\theta) + \epsilon \quad (1)$$

$$y = g(\theta), \quad (2)$$

where ϵ represents an additive measurement noise on the pressure readings.

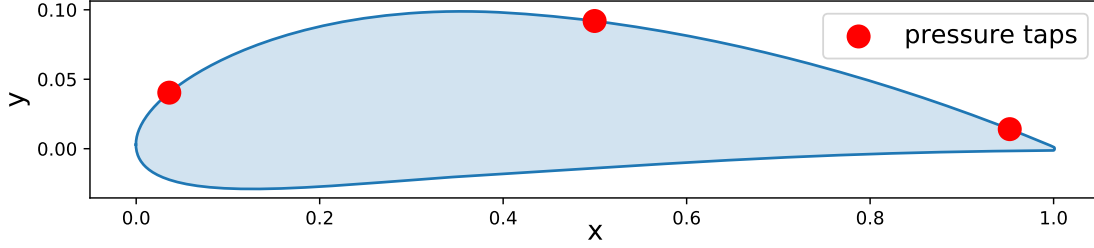
A typical approach to this problem encompasses solving an inverse problem to estimate θ from observed x , and then using this estimated θ to predict the QoIs y . When sparse and noisy data are available, it is often advantageous to solve the inverse problem using a probabilistic framework (i.e., Bayesian inference [13, 14]) that is capable of quantifying the uncertainty in its results. The procedure involves first computing the Bayesian posterior probability density function (PDF) $p(\theta|x)$, and then propagating this θ -PDF through Eq. (2) to arrive at the posterior-predictive PDF $p(y|x)$. This solution path, while mathematically sound, is computationally expensive: characterizing the posterior requires Markov chain Monte Carlo (MCMC) (e.g., [15]) and the subsequent propagation entails Monte Carlo (MC) sampling—the number of forward solves (i.e., CFD simulations) can easily surpass $O(10^5)$ or higher just to process one set of pressure readings, and therefore prohibitive for real-time estimation tasks.

We take a different approach, and seek to construct a mapping directly from x to y while bypassing θ altogether:

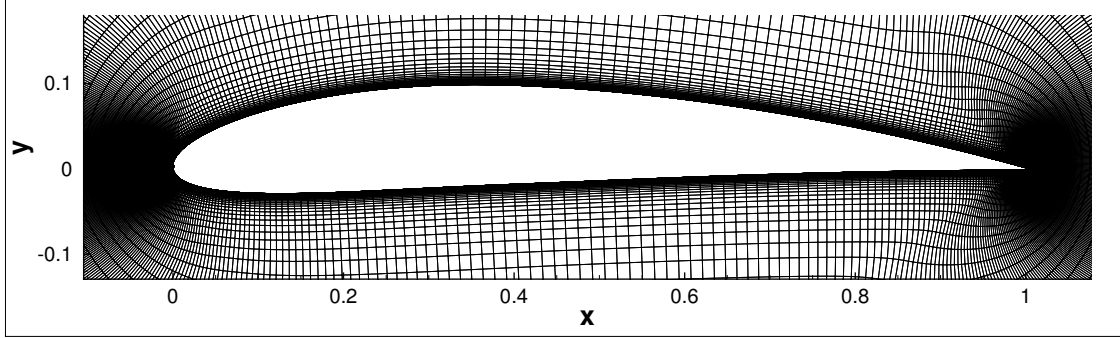
$$\{c_{p,i}\}_{i=1}^{n_p} \longrightarrow \{C_L, C_D, C_{M_z}, x_s\}. \quad (3)$$

Furthermore, we will use (x, y) data pairs generated from the CFD solver, and adopt a supervised learning approach to train data-driven models h based on the relationship

$$y = h(x; w) + \eta, \quad (4)$$



(a) NACA4421 airfoil and sample pressure tap locations



(b) Computational mesh

Fig. 1 Figure (a) shows the airfoil with sample pressure taps located on the upper surface and represented by the red dots. Figure (b) provides a zoomed-in view of the computational mesh near the airfoil.

where w represents the tunable model parameters and η is a statistical representation of the discrepancy between the model predictions and true labels y . Once h is constructed and available, we can directly input pressure readings x to obtain QoI predictions. Evaluations of the data-driven model h are usually extremely fast, thus offering a promising path towards online and real-time usage.

Our second goal is to assess the ML predictive performance under different pressure tap configurations and to find the optimal designs. To achieve this, we can repeat the above procedure and build h for different number of pressure taps and also with varying placement locations, and assess their capabilities under an uncertain environment induced by the operating conditions, measurement noise, and training data.

III. Methodology

A. Data Generation

The dataset is created by performing CFD simulations for a NACA4412 airfoil under various flight conditions θ . To ensure sample diversity and flexibility for growing the dataset in the future, we perform randomized uniform sampling of θ across pre-selected ranges shown in Table 1. For each sample, we solve the two-dimensional compressible RANS equations with the Spalart-Allmaras (SA) turbulence model in steady state using SU2 [16], an open-source software in C++ and Python created for multi-physics simulation and design. The solution is obtained via a finite volume discretization of 60398 elements (Fig. 1(b)) on a mesh that extends 20 chord lengths into the far field, and converged to steady state in pseudo-time using a second-order accurate scheme in space. Upon completion of each run, pressure coefficient values can be extracted at the desired surface pressure tap locations, thus giving our values of x . Aerodynamic performance QoIs are also directly accessible, giving us the corresponding values of y . In this work, the separation location x_s is identified as the most upstream location on the upper surface where the skin friction coefficient switches from positive to negative. Each run takes about 20 minutes with 4 CPU cores. Preliminarily, we have established a dataset of 214 simulations.

Table 1 Operating condition ranges

	Range	Description
M_∞	[0.15, 0.9]	Free-stream Mach number
Re	$[3, 9] \times 10^6$	Reynolds number
α	$[0, 15]^\circ$	Angle of attack

B. Bayesian Neural Networks

A NN maps input x to output $\hat{y}$: we write $\hat{y} = h(x)$ where $\hat{y}$ (with hat) denotes the NN *prediction*. For simplicity, we focus on densely connected feed-forward NNs in this paper. For example, Fig. 2(a) shows a graphical representation of such a NN with an input layer, two hidden layers, and an output layer. A good choice of NN hyperparameters—that is, the number of layers, number of nodes in each layer, and activation functions—often relies on user experience and trial-and-error. We do not conduct a systematic hyperparameter selection in this work, and directly impose reasonable choices. Once the network architecture is decided, the NN can be trained by tuning its weight parameters (we denote them collectively by w). Accordingly, we update our notation to $\hat{y} = h(x; w)$, matching that of Eq. (4) (and so $y = \hat{y} + \eta$).

Given N training data points in the form of input-output pairs $\mathcal{D} := \{x_n, y_n\}_{n=1}^N$, NN training typically involves choosing w to minimize the least squares loss function:

$$w^* = \underset{w}{\operatorname{argmin}} \left[\frac{1}{N} \sum_{n=1}^N \|h(x_n; w) - y_n\|^2 \right]. \quad (5)$$

The optimization is often carried out with stochastic gradient descent [17, 18]. However, regardless of how Eq. (5) is solved, its solution returns a single-valued w^* . Consequently, the prediction at a given input x is also single-valued: $\hat{y} = h(x; w^*)$, and it does not carry any uncertainty information that is expected to be affected by the quantity and quality of the available training data.

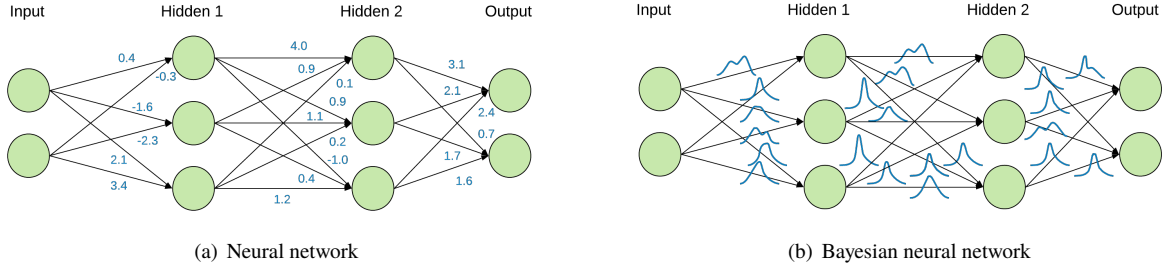


Fig. 2 NN and BNN, each with an input layer, two hidden layers, and an output layer. Weights are shown in blue as deterministic values in NN, and as probability density functions in BNN.

To capture the uncertainty induced by the training data, we employ BNNs. BNNs are similar to NN but treat w as random variables with associated PDFs, as depicted in Fig. 2(b). When training data become available, these PDFs are updated through Bayes' rule:

$$p(w|\mathcal{D}) = \frac{p(\mathcal{D}|w)p(w)}{p(\mathcal{D})}, \quad (6)$$

where $p(w)$ is the prior PDF of w , $p(\mathcal{D}|w)$ is the likelihood function, $p(\mathcal{D})$ is the Bayesian evidence, and $p(w|\mathcal{D})$ is the posterior PDF of w (i.e., when conditioned on training data). Constructing a BNN thus involves computing the posterior $p(w|\mathcal{D})$. However, it can be extremely expensive or even impractical to characterize the full posterior $p(w|\mathcal{D})$ in a high dimensional w -space, with the number of weights found in NNs employed by engineering and science applications often reaching thousands or even millions. Traditional MCMC methods for performing Bayesian inference quickly become infeasible, and approaches with additional simplifying strategies are needed. For example, mean-field variational inference (VI) [19] is a powerful approach for tackling the high-dimensional Bayesian inference task in BNNs [11], based on the idea of seeking the best posterior PDF approximation from a family of independent distributions that are

convenient to work with (e.g., independent Gaussians). The detail of our mean-field VI BNN implementation is omitted in this paper, but can be found in [20] and a companion paper in this conference [21].

Once the BNN is constructed and its weight posterior $p(w|\mathcal{D})$ approximated, we can propagate this weight uncertainty to the predictive QoIs at a given input x using MC: sample different w values from $p(w|\mathcal{D})$ and for each sample evaluate $\hat{y} = h(x; w)$. These $\hat{y}$ samples then correspond to the posterior-predictive PDF $p(\hat{y}|x, \mathcal{D})$. This final result is thus a distribution of QoI predictions (in contrast to the single-value prediction from the NN case), reflecting the uncertainty induced by the training data.

C. Performance Assessment

In order to compare the performance of different pressure tap designs, we propose to use the following loss function:

$$L = \mathbb{E}_{\theta} \left[\mathbb{E}_{x|f(\theta)} \left[\mathbb{E}_{\hat{y}|x, \mathcal{D}} \left[\|\hat{y} - g(\theta)\|^2 \right] \right] \right]. \quad (7)$$

We explain each component of this criterion starting from the innermost term, and proceed outwards.

- $\|\hat{y} - g(\theta)\|^2$: At a given test data point (i.e., a test flight condition), we can compute the squared error between the NN prediction $\hat{y} = h(x; w)$ and the true QoI values $g(\theta)$.
- $\mathbb{E}_{\hat{y}|x, \mathcal{D}}$: However, $\hat{y}$ is a random variable in the BNN setting, since it inherits the uncertainty from w that was originally induced by the training data. Therefore, we take the expectation over the posterior-predictive PDF $p(\hat{y}|x, \mathcal{D})$, to obtain the squared error averaged over the entire distribution of QoI predictions. In practice, BNN will generate a group of prediction samples following the posterior-predictive PDF, such as shown in Fig. 3. For each prediction sample, we can compute the squared error, and then take the average error from all samples.
- $\mathbb{E}_{x|f(\theta)}$: Here we take into account of the fact that pressure tap measurements are noisy, as expressed by Eq. (1), and therefore take the expectation over the noisy measurement distribution. We note that this effect is not present in the BNN training stage since the training data are noise-free; it is only included in this performance loss in anticipation of noisy measurements when the BNN is deployed and used in the real environment.
- $\mathbb{E}_{\theta}$: The last expectation is to account for different possible operating conditions θ , following the uniform distribution described in Table 1.

Overall, this loss function represents the expected predictive error, where the expectations reflect the uncertainty in operating conditions, pressure measurement noise, and training data. It can be approximated numerically using the following MC estimator:

$$L \approx \frac{1}{n_{\theta}} \sum_{i=1}^{n_{\theta}} \left[\frac{1}{n_{\epsilon}} \sum_{j=1}^{n_{\epsilon}} \left(\frac{1}{n_w} \sum_{k=1}^{n_w} \|\hat{y}_{i,j,k} - g(\theta_i)\|^2 \right) \right], \quad (8)$$

where we have adopted $n_{\theta} = 43$, $n_{\epsilon} = 100$, and $n_w = 1000$ respectively for the number of test data points (test flight conditions), measurement noise samples, and BNN weight samples. We can also decompose the total loss to individual contributions from each of the four QoIs by applying the same formula to the appropriate individual QoI component.

IV. Results

We present results for one-, two-, and three-tap designs ($n_p = 1, 2, 3$). For each design configuration, a separate BNN is constructed. To set up the BNN problem in Eq. (6), the prior $p(w)$ is chosen to be independent Gaussian $w_d \sim \mathcal{N}(0, 1^2)$ for each dimension of w , and the likelihood $p(\mathcal{D}|w)$ reflects the additive independent Gaussian noise as presented in Eq. (4) with $\eta^{(i)} \sim \mathcal{N}\left(0, \sigma_{\eta^{(i)}}^2\right)$, where $\eta^{(i)}$ denotes the i -th component in the noise term η , and $\sigma_{\eta^{(i)}}$ is chosen to be 20% of the standard deviation of the i -th QoI among all data points in the training set. Table 2 summarizes the BNN architecture adopted in our investigation, leading to a total of $50n_p + 254$ network weights corresponding to $100n_p + 508$ variational parameters (each weight w_d has a posterior μ_d and σ_d). A total of 214 CFD simulations is currently available, and 80% of them (171 runs) are used for training while the rest (43 runs) are held for testing. Batch gradient descent is employed through the ADAM optimization algorithm [22], with learning rate set to 0.3 and training up to 10,000 epochs.

A. One-Tap Design

We begin by investigating the predictive performance of BNN for only one pressure tap; the design variable in this case is then the x -coordinate of the sole sensor (i.e., x_1). The design space is discretized in accordance to the 300

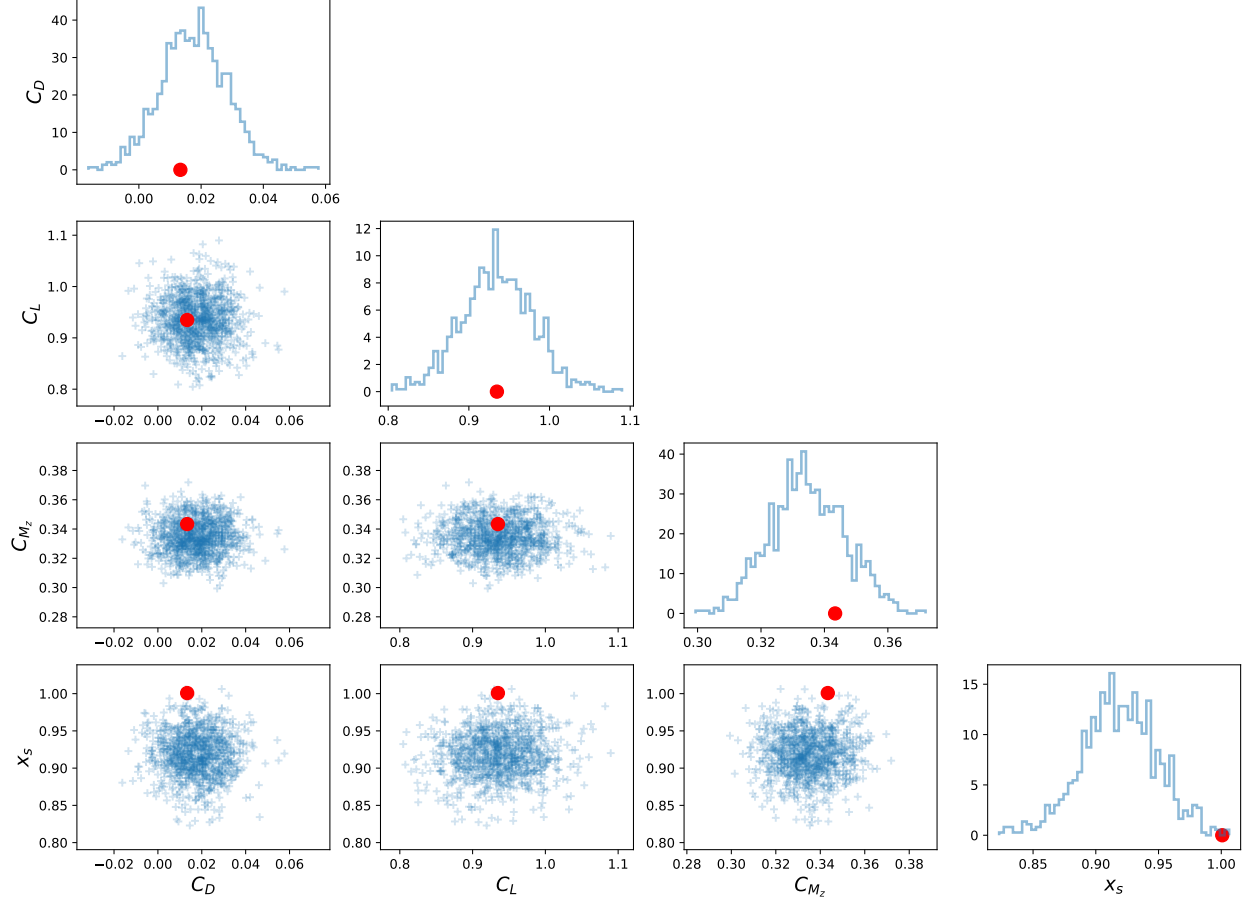


Fig. 3 BNN prediction samples under the first testing flight condition for the optimal three-tap design (described in section IV.C). Blue crosses (off-diagonal sub-plots) and blue lines (on-diagonal sub-plots) represent the prediction samples, and red dots indicate the true output QoI values.

computational grid points on the airfoil upper surface. At each grid point, the c_p values from the 214 CFD training points are extracted to construct a BNN to predict the QoIs. The BNN performance is then assessed using Eq. (8) on the 43 test points. The measurement noise in Eq. (1) is chosen to be $\epsilon = \epsilon_1 + \epsilon_2$, where $\epsilon_1 \sim (0, \sigma_{\epsilon_1}^2)$ is a base term with σ_{ϵ_1} fixed at 5% of the standard deviation of the c_p values among the 271 training samples, and $\epsilon_2 \sim (0, \sigma_{\epsilon_2}^2)$ is a relative term with σ_{ϵ_2} proportional to the c_p magnitude. We compare results using σ_{ϵ_2} at noise levels of 0%, 1%, 5%, 10%, 20% and 50% of the c_p magnitude.

Figure 4 presents the performance loss associated with each QoI under various pressure tap locations and σ_{ϵ_2} noise levels, and Fig. 5 shows the total loss (sum of individual losses from all four QoIs). Observing Fig. 4, it is clear that C_L and x_s dominate the loss contributions. Furthermore, when the noise level is 20% or less, x_s dominates the loss if the pressure tap is in the front of the airfoil, and C_L dominates if the tap is in the back—the optimal tap location thus changes depending on which QoI is targeted. Contributions from C_D and C_{M_z} are negligible in comparison, which is reasonable since their variations are much lower across different flight conditions in this study. The loss characteristics of each QoI remains nearly unchanged when the noise level is at or below 10%, which indicates that the trained BNNs are robust under 10% pressure measurement noise. However, when the noise increases to 50%, the prediction of C_L collapses, especially when the tap is placed near the middle of the domain (note the y-axis scale in the plots). From Fig. 5, we find that the overall optimal design location is either just behind the leading edge or near the trailing edge.

To extract further insights of the predictive challenges faced by the BNNs, we plot the input-output responses that the BNNs are trying to capture: Fig. 6 presents the testing and predicted QoI values (figure y-axis) as responses of pressure readings (figure x-axis) under various pressure tap locations (each column) and for each of the four QoIs (each row). The

Table 2 BNN architecture

Layer	Name	Number of Neurons	Activation Function
1	input	n_p	-
2	hidden 1	50	tanh
3	output	4	linear

blue dots represent the testing data points (training data have similar distributions and trends and thus omitted), and the black dotted lines are the BNN mean predictions accompanied by the orange $\pm 2\sigma$ uncertainty intervals. Studying these figures, when the pressure tap is near the leading edge or the trailing edge, the relationship between an QoI output to the c_p input appears nearly one-to-one for C_D , C_L and C_{M_z} and thus relatively easy for BNNs to capture; however, when the tap is near the middle of the airfoil, especially at $x_1 = 0.6$ (fourth column), the responses become visibly multi-valued and therefore very difficult for any functional approximator (such as NNs) to represent. Physically, the reason behind the effectiveness of the leading or trailing edge pressure taps is due to the fact that any variation of operating conditions results in immediate change of the leading and/or trailing edge stagnation points. For x_s , the responses are always multi-valued, but the trailing edge design still offers a relatively benign trend for the BNNs to capture. In all the plots, the predictive uncertainty (orange envelope) is large where the data is sparse and small where the data is dense, which is useful for indicating the uncertainty of BNN predictions induced by the availability and location of training data points. However, they are currently unable to capture the full range of data variability, likely due to model inadequacy of the current BNN architecture and the assumed likelihood structure and data noise distributions. Overall, these challenges of multi-valued responses are caused by the solution non-uniqueness of the underlying inverse problem (that is, when only a single pressure tap is available at a fixed location, there may be multiple flight conditions that produce the same c_p observation), subsequently leading the BNNs to have higher predictive errors near the middle of the airfoil.

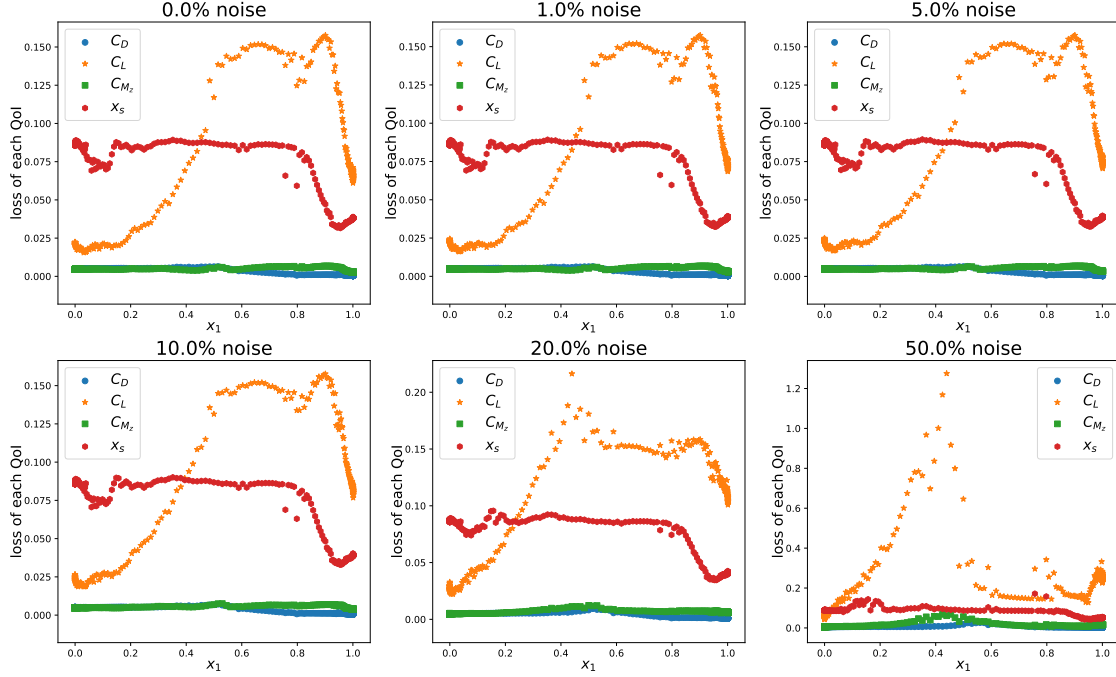


Fig. 4 Performance loss function of each QoI of one-tap design under various sensor locations and noise levels.

B. Two-Tap Design

For a two-tap configuration, the design variables are the x -coordinates of the two sensors (i.e., x_1 and x_2). An exhaustive exploration of all possible design combinations using the 300-node grid would be 300^2 , a task that is

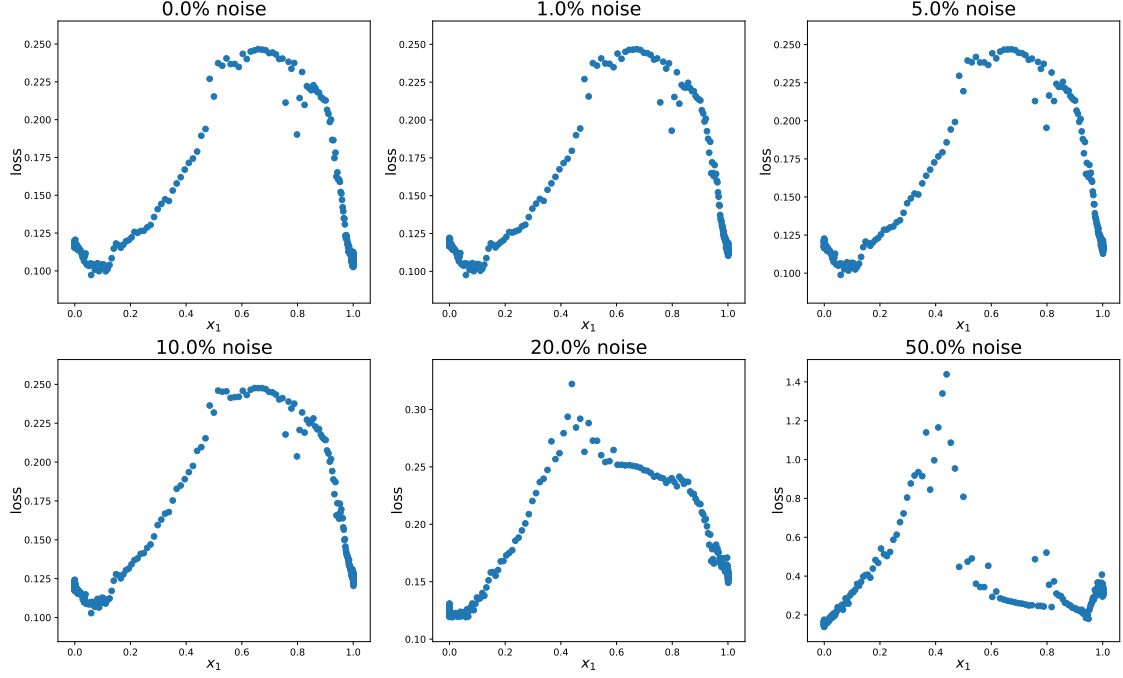


Fig. 5 Performance loss function of one-tap design under various sensor locations and noise levels.

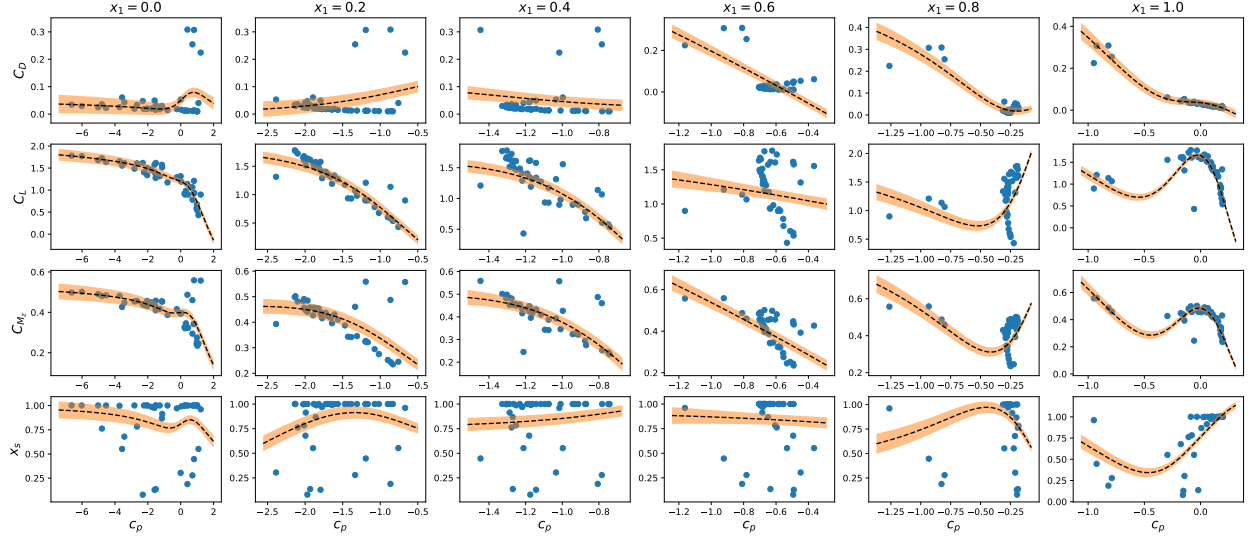


Fig. 6 Testing and predicted QoI values (with no pressure measurement noise) plotted against pressure readings at various sensor locations. Blue dots indicate testing data points, black dashed lines are the predictive mean, and orange envelopes are $\pm 2\sigma$ predictive uncertainty.

computationally prohibitive. Instead, we create a smaller design space by retaining only 11 of the original grid points that are nearly uniformly distributed between the leading edge and trailing edge as candidate sensor locations, and investigate the performance of BNNs on those 11^2 two-tap configurations. Figure 7 presents the performance loss heatmaps under different configurations and noise levels, and Fig. 8 shows the optimal two-tap designs. The performance is generally higher when the two taps are far away from each other, and lower when close together. The optimal design combination always entails one sensor at the leading edge and the other at the trailing edge when the noise level is 20% or less. The loss function also remains similar from 0% to 10% noise levels, again suggesting robustness in the BNNs.

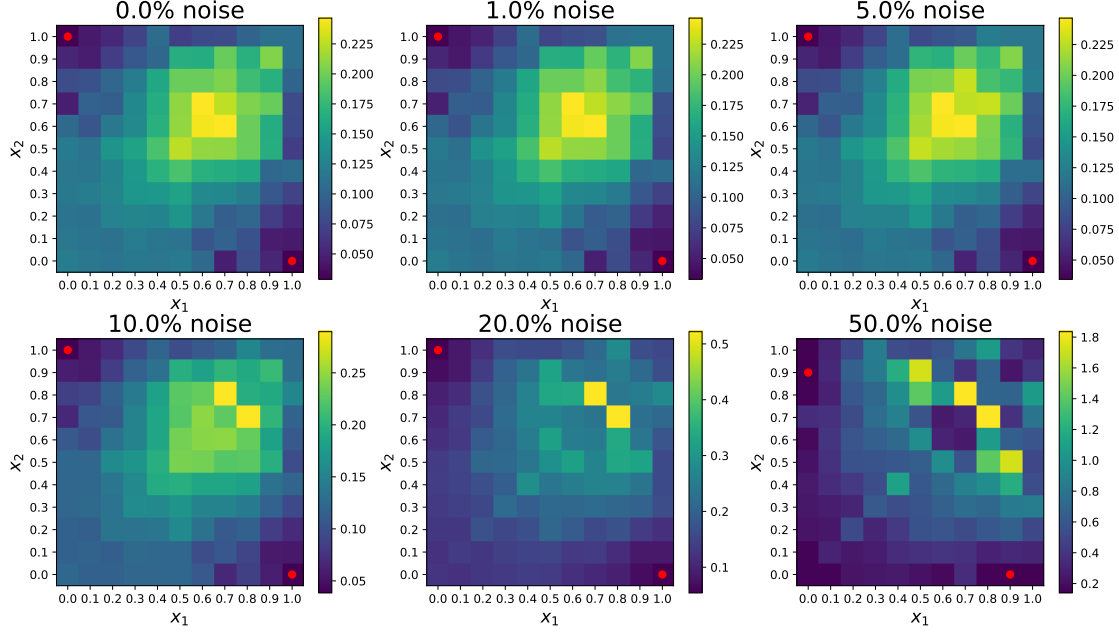


Fig. 7 Heatmaps of performance loss function of two-tap design under various locations and noise levels. Red dots indicate the optimal design locations.

Figure 9 displays the input-output responses the BNNs are trying to capture, plotting the testing and predicted QoIs as responses of c_p readings. We observe that when two taps are close ($x_1 = 0.4$ and $x_2 = 0.6$), QoI responses are again multi-valued, and the BNNs can only capture the mean of these multiple values while a high number of testing points remain outside the $\pm 2\sigma$ envelope. However, when two taps are far away from each other ($x_1 = 0.0$ and $x_2 = 1.0$), the response is almost one-to-one, and almost all of the testing points are within the $\pm 2\sigma$ envelope. Comparing Fig. 9 with Fig 6, we see that the extent of multi-valued-ness is less severe in the two-tap design as a result of “lifting” the surface into a higher-dimensional space.

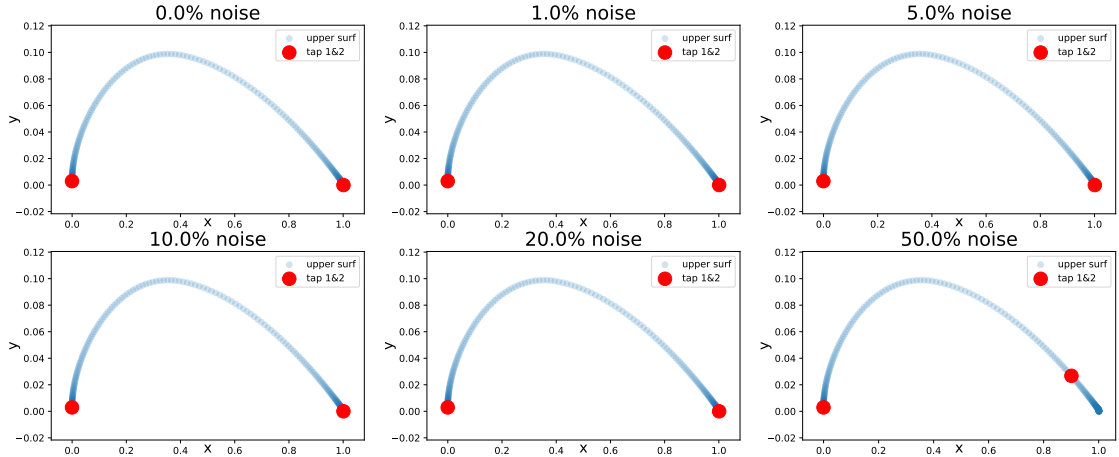


Fig. 8 Optimal locations of two-tap design under various noise levels.

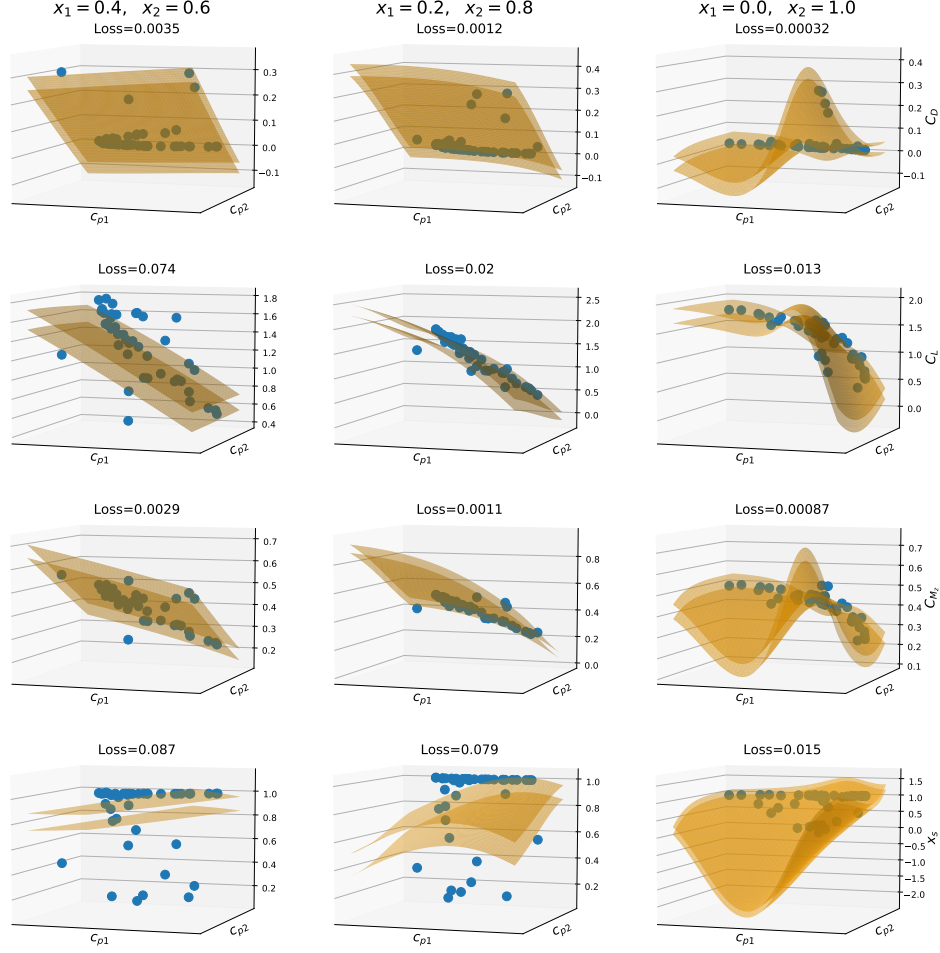


Fig. 9 Testing and predicted QoI values (with no pressure measurement noise) plotted against two-tap pressure readings from various combinations of sensor locations. Blue dots indicate testing data points, black dashed lines are the predictive mean, and orange envelopes are $\pm 2\sigma$ predictive uncertainty.

C. Three-Tap Design

Lastly, we explore the design of three taps with a similar setup as the two-tap case, by choosing x_1 , x_2 and x_3 out of 11 possible grid location for a total of 11^3 combinations. While we have conducted a full design space search, we show in Fig. 10 the heatmaps of performance loss when the first tap is fixed at the leading edge (i.e., $x_1 = 0$). Under this setting, the performance deteriorates when the other two sensors are both near the leading edge, in which all three sensors are clustered together and do not provide a diversity of information. Figure 11 shows the optimal locations under various noise levels. When the noise is small, it is best for the first sensor to be located at the leading edge, the second slightly behind, and the third at the trailing edge. However, we notice that the location of the second sensor is not very sensitive if we already have two sensors at the two end points, since the loss values of the top row (or rightmost column) in the individual heatmaps are fairly flat. When the noise level increases to 50%, the heatmap appears similar to that of the two-tap design, which indicates that it is better to have another sensor also located at the leading edge to reduce the impact of high measurement noise level, such that the optimal locations are two taps at the leading edge and one tap at the trailing edge. Comparing across the one-, two-, and three-tap designs, the performance loss values decrease overall as the sensor number increases, a reasonable behavior since one expects more sensors to provide more information.

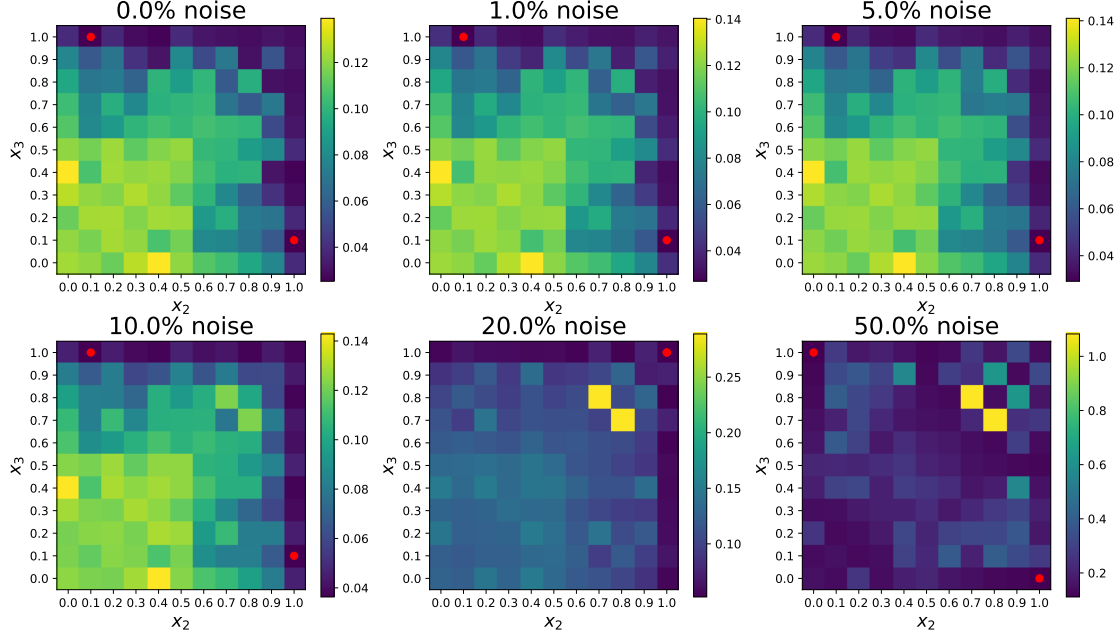


Fig. 10 Heatmaps of performance loss function of three-tap design under various locations and noise levels, and with the first tap fixed at the leading edge ($x_1 = 0$). Red dots indicate the optimal design locations.

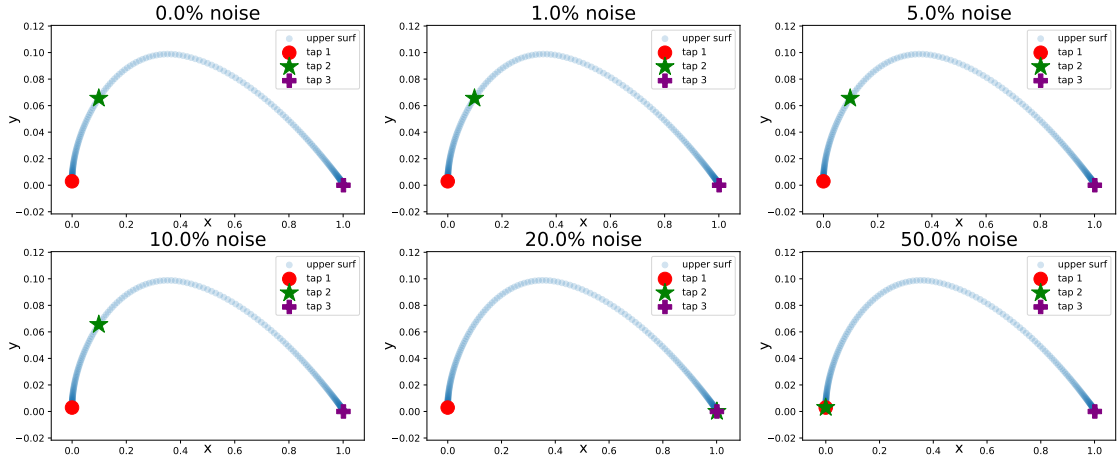


Fig. 11 Optimal locations of three-tap design under various noise levels.

V. Conclusions

In this work, we proposed a ML-based tool to aid the design of airfoil pressure tap locations for achieving real-time predictions of aerodynamic QoIs under noisy and sparse measurements and for eventually supporting active flow separation control technologies. These tools consisted of BNNs, which were constructed in a supervised learning manner using a database of SU2 CFD simulations of flow around a NACA4412 airfoil under various flight conditions. The BNNs were constructed offline on the ground leveraging high-performance computing resources and physics-based simulations, and then utilized for rapid online predictions in-flight. The evaluation speed of a BNN is extremely fast, since it bypasses the expensive inverse problem and establishes a direct mapping from inputs of sensor pressure readings to predictions of the lift, drag, and moment coefficients and flow separation location. Furthermore, BNNs are attractive for incorporating uncertainty quantification in ML: they output not only single-valued prediction quantities, but also the

uncertainty information surrounding these predictions, thus providing a sense of confidence of the ML computations. This predictive uncertainty is generally never reported in traditional NN methods, and is crucial for decision making and controller design that concern the safety of the aircraft and its passengers.

We investigated one-, two- and three-tap designs of pressure sensor locations, where the optimal configurations minimized the expected prediction error at testing data points. We observed better performance when pressure sensors were placed close to the leading edge and/or the trailing edge, but not clustered together for the multi-tap cases. In the future, we will explore higher fidelity physics-based simulations for training the ML models using three-dimensional flows and large eddy simulations, and investigate the performance behavior with higher number of sensors.

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